

SANDEEP ARLEE

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LinkedIn — GitHub — Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer and Full Stack Developer with experience building healthcare AI systems, scalable backend APIs, and production-ready web applications. Skilled in Python, TensorFlow, FastAPI, React, TypeScript, PostgreSQL, and AWS-based deployment workflows. Experienced in deep learning pipelines, backend integration, REST API development, cloud deployment, and frontend optimization with a strong focus on scalability, performance, and maintainable architecture.

PROFESSIONAL EXPERIENCE

AI/ML Engineer & Full Stack Developer

June 2025 – Present

Griva Healthcare Pvt. Ltd., Visakhapatnam

- Built and optimized an EfficientNet-B3 based cervical cancer detection pipeline using transfer learning, image preprocessing, and augmentation techniques for medical image classification.
- Worked on deployment workflows, PostgreSQL database integration, AWS EC2 infrastructure, and production backend configuration for scalable healthcare platforms.
- Contributed to frontend application development using React, TypeScript, and Tailwind CSS with focus on responsive UI and performance optimization.

Machine Learning Developer

June 2023 – May 2024

SS Institution, Visakhapatnam

- Designed, trained, and deployed machine learning models for object detection, fake audio detection, and price prediction using TensorFlow, Scikit-learn, OpenCV, and Python.
- Performed data preprocessing, feature engineering, dataset balancing, and model evaluation to improve prediction accuracy and inference performance.
- Integrated trained ML models into Python applications and automated prediction workflows for real-world use cases.
- Worked with NumPy, Pandas, and OpenCV for data processing, image analysis, and model pipeline optimization.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, JavaScript, TypeScript
- **Machine Learning & AI:** TensorFlow, Scikit-learn, OpenCV, EfficientNet, Deep Learning, Transfer Learning, CNNs, Data Preprocessing, Model Evaluation
- **Backend Development:** FastAPI, Flask, Node.js, REST APIs, API Integration, Authentication, Backend Architecture
- **Frontend Development:** React, Vite, HTML5, CSS3, Tailwind CSS, Responsive UI Development
- **Databases:** PostgreSQL, MongoDB, MySQL
- **Cloud & DevOps:** AWS EC2, Git, GitHub, Linux, Deployment Workflows
- **Tools & Platforms:** Firebase, Razorpay Payment Gateway, Google Maps API, GitHub Pages

PROJECTS

GRIVA Healthcare Platform

React, TypeScript, AWS, PostgreSQL

- Built and deployed a production-grade healthcare platform with responsive frontend architecture using React, TypeScript, and Tailwind CSS.
- Integrated backend APIs, dynamic blog systems, SEO optimization, and Google Maps API functionality.
- Contributed to backend deployment workflows, infrastructure updates, and production environment configuration.

AI-Based Cervical Cancer Screening System

Python, TensorFlow, EfficientNet-B3, OpenCV

- Developed a deep learning-based cervical cancer screening system using EfficientNet-B3 to classify cervical images and assist early detection workflows.
- Designed an image preprocessing and inference pipeline incorporating feature extraction, data augmentation, and model optimization techniques for medical image analysis.

Project Mentor Hub

React, Razorpay, JavaScript

- Developed an academic project marketplace with Razorpay payment gateway integration and responsive UI workflows.
- Implemented project browsing, payment workflows, and reusable frontend components for improved user experience.

EDUCATION

Master of Computer Applications (MCA)

June 2024 – May 2026

VIT Bhopal University, Bhopal, Madhya Pradesh

CGPA: 8.0 / 10

Bachelor of Computer Applications (BCA)

June 2020 – May 2023

Gayatri Vidya Parishad, Visakhapatnam, Andhra Pradesh

CGPA: 7.0 / 10